DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
BANGLADESH UNIVERSITY OF BUSINESS AND TECHNOLOGY

COURSE OUTLINE

1	**Faculty**	Faculty of Engineering & Applied Sciences (FEAS)
2	**Department**	Department of Computer Science and Engineering
3	**Program**	B.Sc. Engg. in Computer Science and Engineering (B. Sc. Engg. in CSE)
4	**Course Code**	CSE 221
5	**Course Title**	Data Structure
6	**Course Type**	Core Engineering Course
7	**Pre-requisites**	CSE 101
8	**Credit Hours/Value**	1.5
9	**Contact Hours**	17 weeks × 1 classes per week × 2.5 hours per class = 42.5 total hours
10	**Year- Semester**	2 - 1
11	**Academic Session**	Spring 2025

12	**Class Schedule**	**Intake – Section (Shift)**	**Class Day**	**Class Hours**	**Venue**
		53–1 (Day)	Monday	9:15 AM – 10:30 AM	B2/319
			Wednesday	5:15 PM – 6:30 PM	B2/905
		BUBT Campus, Rupnagar, Mirpur 2, Dhaka - 1216			

13	**Course Website**	Google Classroom Code: 24vlecv

14	**Course Teacher's Information**	**Name (Code):** Nasrin Akter	
		Designation: Lecturer Department of CSE	**Email:** nasrin.akter@bubt.edu.bd **Cell No.** +8801837287715

15	**Counselling Hour/Tutorial**	**Day**	**Counseling Hours**	**Venue**
		Monday	10:30 AM-11:45 AM	B1/602
		Wednesday	05:15 PM -06:30 PM	

16	**Accessories & Aids**	Students must carry learning materials like lecture notes, calculator, pen, pencil, eraser etc. in the classroom. Borrowing learning materials in the classroom or exam room from fellow students is prohibited. A student is also advised to keep a separate class note (khata) of 50 pages for the course during class hours.
17	**Course Rationale**	This sessional (lab) course is based on the theory course CSE-221 Data Structures. The course is designed for students to perform various data structure operations and their applications. The student will learn to select appropriate data structure according to engineering problem. This course enables students to get acquainted with the concepts of time-space complexity.
18	**Course Objective**	This course equips students with essential skills in designing, analyzing, and implementing diverse data structures. Through hands-on exercises, students will learn to perform data structure operations and apply them to solve engineering problems. The course emphasizes the selection of appropriate data structures based on specific needs and provides insights into time and space complexity. By the end, students will be well-prepared to make informed decisions in real-world engineering scenarios, contributing to the development of efficient solutions.
19	**Course Learning Outcomes (CLOs)**	Upon completing this course student will be able to: **CLO1: Understand** the basic data structures and their area of usage. **CLO2: Implement** the operations and manipulations of different data structures and their memory organization.

		CLO3: Apply appropriate data structures for solving computational problems.

20 Mapping of CO – PO Method of Delivery and Assessment Tool

CLOs	PLOs &CF	Bloom's Domain / Level	Delivery Methods / Activities	Assessment Tools
CLO1	PLO1 CF = 2	Cognitive/Understanding	⦿ Classroom Lecture (PPP &/or WBT*) ⦿ Classroom Discussion& Exercise practice ⦿ Analysis and design problem solving ⦿ Course Counseling	***Indirect*: In class response, course counseling, course end survey ***Direct*: Class participation and activity, class test, assignment, midterm and final examinations
CLO2	PLO1 CF = 3	Cognitive/Understanding		
CLO3	PLO2 CF = 3	Cognitive/Analyzing		

*PPP: Power Point Presentation & WBL: White Board Teaching

Correlation of COs to POs

Correlation Criteria	Correlation Level	Correlation Factor (CF)
Less than 25%	Almost no correlation	0
Equal to or greater than 25% and below50%	Poor (Low)	1
Equal to or greater than 50% and below 75%	Moderate (Medium)	2
Equal to or greater than 75%and up to 100%	Significant (High)	3

21 Teaching-Learning Strategy

The course's teaching-learning process is designed to achieve its intended learning outcomes. Various classroom tools, such as multimedia projectors with desktop computers, whiteboards, and overhead projectors, are used to make the process engaging, effective, and comprehensive. The primary method of teaching is through classroom lectures, wherein most of the course content is covered in the lecture notes. For the remaining topics, textbooks are utilized, with additional references provided for students to study on their own. Lecture materials are posted on Google Classroom web pages and also provided as hard copies in the classroom. To ensure that students achieve the expected performance and knowledge level, classroom discussions, PowerPoint presentations, problem-solving using whiteboard markers, and homework or home studies are used. Counseling is also offered to help students with weak areas. Formative assessments of individual students are done through inside and outside classroom discussions, in-class eye contact and clicker questions, homework, and students' responses. A course-end survey is also conducted. Summative assessments are done through class participation and performance observation, assignments, class tests, and semester midterm and final examinations.

If a student is absent from a class for any reason, they are advised to do self-study and take tutorials from the class teacher to make up for the missed class. Supplementary examinations are available for students who missed the midterm or final examinations due to valid reasons. These supplementary exams are more challenging than the regular exams.

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Course Plan

This course consists of 150.0 min./week of class contact hours and an additional 90 min./week of counseling hours to explain students' design problems, provide reading materials, and assist in understanding lecture materials for preparing examinations, class tests, and assignments.

Week	Selected Topics	Lecture Note, Text Book& Other Ref.	Teaching Learning Strategy	Assessment Strategy	Correspon ding CLOs
1-2 (Week 1)	Introduction to Data, Information, Structures, Data Structure Operations, ADT, T-S Trade-off, Notation & functions, Control Structures, Algorithm & Program Design.	Data Structures (Schaum's Outlines Series) (Chap-01)	Lecture, Video Presentation		CO1
3-4 (Week 2)	Big Oh Notation Discussion, Sub algorithm, local & global concepts, Algorithm & Program Design, Complexity Analysis.	Chap-01 & Chap-02			CO1
5-6 (Week 3)	String, String Operations, Word processing, First Pattern Matching Algorithm, Second Pattern Matching Algorithm.	Chap-03			CO2, CO3
7-8 (Week 4)	Array, Array Operations (Insertion, Deletion) Array Operations (Sorting, Searching, Bubble Sort), Linear Search & complexity, Binary Search, Pointers & complexity, Matrices and their Operations.	Chap-04	Lecture discussion with White Board and Multimedia, Problem Solving	Class Test, Assignment, Written Examination	CO2
9-10 (Week 5)	Matrices and their operations (Cont.), Selection sort and complexity, Insertion Sort & complexity, Radix Sort & Complexity, Merge Sort & Complexity.	Chap-04 & Chap-09			CO2, CO3
11-12 (Week 6)	Introduction to Linked List, Usage of linked list, Create Linked Lists etc. Linked list Operations (Insertion, Deletion, traversing etc.), Dynamic Memory Allocation, Linked list Vs. Array Representation.	Chap-05			CO2
13-14 (Week 7)	Case Study & Review class for Mid Term Examination.		Lectures, Questions and answers	Problem Solving & Viva Voce	CO2
Midterm Examination (10 Mar – 25 Mar 2025)					
15-16					

(Week 8)	~~Hashing, Collision resolution, Methods~~ Stacks and its operations).	Chap-09			CO2, CO3
17-18 (Week 9)	Applications of Stack: Polish Notation, Polish Notations using stack.	Chap-06			CO3
19-20 (Week 10)	Recursion, Queues & Applications, Deques, Priority Queues, Tower of Hanoi.	Chap-06			CO3
21-22 (Week 11)	Ackerman function, Tree's terms & Definition, Types of tree, B-Trees.	Chap-06	Lecture discussion with White Board and Multimedia, Problem Solving	Class Test, Assignment, Problem Solving Ability and Written Examination	CO2
23-24 (Week 12)	Tree traversing order, Binary Search-Trees (Insert, delete).	Chap-07			CO2
25-26 (Week 13)	Heap(Min-Max), Huffman Coding, AVL Tree.	Chap-07			CO2, CO4
27-28 (Week 14)	Graph terms, Adjacency matrix, Path matrix, POSET, BFS.	Chap-08	Discussion and Problem Solving	Problem Solving Ability and Written Examination	CO2
29-30 (Week 15)	DFS, Data Structure Problem Analysis, modification & Related Solution methods.	Chap-08			CO4
31-32 (Week 16)	**Brain Storming Week**				
33-34 (Week 17)	Case Study & Review class for Semester Final Examination.		Lectures, Questions and answers	Problem Solving & Viva Voce	CO4
	Final Exam (18 June -2 July 2025)				

23 i	**Text Books**	1. Data Structures (Schaum's Outlines Series) - Seymour Lipschutz
ii	**Reference Books**	1. Data Structures Using C ISRD Group.
		2. Data Structures and Algorithms. Aho, Ullman & Hopcroft.

24 Assessment and marks distribution criteria

Active engagement in class activities, participation in outside classroom discussions, and communication through the Internet and phone are integral parts of this course. Failure to participate in class regularly, take class tests, and/or complete assignments may result in failing the course. To achieve the course-specific expectations, students must actively participate in classroom discussions and complete all sets of work at a satisfactory level, as outlined in the course content. The course-specific expectations for students are achieved if

- 75% of students in a section attend more than 70% of classes (determined by summative assessment).
- Their active participation in the classroom discussion is targeted at up to 80% of the total attendees (determined by formative assessments such as eye contact, clicker questions, and group discussions).
- Equal or more than 40% of course outcomes must be achieved by the students (summative assessment).
- The level of engagement in the studies, such as regularly preparing class lectures, class tests, and assignments, must be more than 60% (formative assessment).
- Expected level of participation in the outside class discussion (once weak, more than 30% of students in a section) by course counseling and using social media like Google Classroom, email, phone call, etc. (formative assessment).
- Students are assessed according to their individual performance in the examinations, class tests, assignments, and class participation. The final mark calculation and course outcome assessment are done based on the following mark distribution criteria:

Assessment tool	Conducting Number	Mark distribution (%)
Class participation and activity	34	05
Class test	2	7.5 × 2 = 15
Assignments/Report and Presentation	1	5.0 + 5.0 = 10
Midterm examination	1	30
Final examination	1	40
	Total Mark	100

Class participation & activity performance criteria

Performance level	Mark distribution (%)
91% - 100%	05
86% - 90%	04
81% - 86%	03
76% - 80%	02
70% - 75%	01
less than 70%	Not allowed to sit for the final examination.

25 Assessment Strategy

Assessment tool	Content, Length and Criteria	Weight (mark)	Due date
Class test 1	Composed of 3 to 5 short questions. Duration of the class test is 40 minutes, and the assigned full mark is 7.5.	7.5	On the 12th lecture
Class test 2	Composed of 3 to 5 short questions. Duration of the class test is 40 minutes, and the assigned full mark is 7.5.	7.5	On the 25th lecture

	Assignment with a presentation/ Viva Voce	Assignment with a presentation/ Viva Voce will be delivered two weeks after assigning. 5 marks are assigned for the assignment and the other 5 marks are for a presentation/ Viva Voce.	10.0	Presentation will be conducted on the 28th lecture
	Midterm Examination	Midterm Examination is held according to the Academic Calendar and examination schedule after 7 weeks of classes. All three sets of questions are to be answered by the students in one and half-hour exam period. Each question set contains multiple short type questions. The assigned full mark for the 3 sets of questions is 30.	30	After 14th lecture and following the examination schedule of BUBT.
	Final Examination	Final Examination is held after 8 weeks of classes after the midterm examination. All four sets of questions are to be answered by the students in a two-hour exam period. Each question set contains multiple short type questions. The assigned full mark for the 4 sets of question is 40.	40	After 9 weeks of classes afterward the midterm examination and following the examination schedule of BUBT.

26	CLO Assessment Criteria	Assessment of CLOs					
		Assessment Tool	CLO				Mark Allocation
			CLO1	CLO2	CLO3	CLO4	
		Class Participation	-	-	-	-	-
		Class Test	-	-	-	-	-
		Assignment& Presentation	-	-	-	-	-
		Midterm Exam	20	10	0	0	30
		Final Exam	0	20	10	10	40
		Total Mark	20	30	10	10	70

27	Rubrics (Attainment Criteria)	CLOs (Taxonomy domain)	Not attained /Failed (0-39%)	Poor (40%-49%)	Moderate (50%-64%)	Good (65%-79%)	Excellent (80%-100%)
		CLO1 (understand) CLO2 (understand and apply) CLO3 (apply) CLO4 (understand and apply)	The question was answered with serious deficiencies in understanding and explanation. Applicable method was not almost touched.	The question was answered inadequately by touching on the applicable method or without explanations. As a result, a few steps of problem-solving procedures or concepts are not developed properly or are missing.	The question was answered partially correctly by applying the method or concepts asked, but a few important details were missing.	The question was answered correctly but briefly, and missed some portions of the important explanation by applying the required method or concepts.	The question was answered correctly with detailed explanations using the asking method of solving the problem or concepts with adequate explanation.

28	Feedback	All kinds of feedback to the students will be produced within a week after the day of holding a class test and midterm examination. No answer script will be shown for the final examination if it is not challenged by a student. Online and email queries will also be responded to within three days by email.

29	Grading Policy	Letter grades and grade points are used to evaluate the performance of a student in the course:		
		Marks Range	Letter Grade	Grade Point
		80% and above	A+ : A Plus (Outstanding)	4.00

		75% to less than 80%	A	: A regular (Excellent)	3.75
		70% to less than 75%	A-	: A minus (Very good)	3.50
		65% to less than 70%	B+	: B Plus (Good)	3.25
		60% to less than 65%	B	: B regular (Average)	3.00
		55% to less than 60%	B-	: B minus (Below average)	2.75
		50% to less than 55%	C+	: C Plus (Poor)	2.50
		45% to less than 50%	C	: C regular (Very poor)	2.25
		40% to less than 45%	D	: Pass marginally	2.00
		Less than 40%	F	: Fail	0.00
			I	: Incomplete	
			R	: Retake	
			W	: Withdraw	

30	**Additional Course Policies**	Assignment	Late assignment submissions will not be accepted. ***Any kind of copied information without proper citation (i.e., plagiarism) in the assignments or any other work is prohibited and treated as serious academic misconduct, which is prosecuted through the Discipline Committee of BUBT.*** All copied or plagiarized assignments/reports/test papers will be canceled, and the isolated student must prepare a new assignment/project again. **Zero tolerance will be shown in this regard.** Feedback after the submission of long assignments will be provided on hand or by email within two weeks.
		Class Test	Two class tests (CTs) will be conducted for the course. All class tests have equal weightage of 7.5. Both regular and surprise CTs can be conducted.
		Closed book assessments	CTs, midterm and final examinations are closed book assessments. Mobile phone is prohibited in the examination hall. Students are insisted on carrying simple scientific calculators to solve the complex calculations and a wrist-watch to follow time during the exam hours.
		Test Policy	If a student is absent from a class test anyway and does not report to the class teacher personally beforehand, his/her score for that test will be zero. No make- up for the class test will be allowed. No supplementary for midterm and final examinations will be entertained without physical presence and recommendation of the guardian, along with written permission of the department. Supplementary examination questions are much harder than the regular examination questions; therefore, students are discouraged from taking supplementary examinations.

31	**Additional Information**	Academic Calendar Spring 2024: https://www.bubt.edu.bd/Home/page_details/Academic_Calender Academic Rules: https://www.bubt.edu.bd/Home/page_details/Rules_and_Regulations Grading & Evaluation: https://www.bubt.edu.bd/Home/page_details/Evaluation_Grading_System Rules& Regulations: https://www.bubt.edu.bd/Home/page_details/Office_of_the_Registrar

32 Bloom's Taxonomy for Teaching-Learning

Bloom's Taxonomy is a set of three hierarchical models used to classify educational learning objectives into levels of complexity and specific quality. The three taxonomy domains for achieving learning objectives are cognitive, affective, and psychomotor. Cognitive domain is in the primary focus of educating and frequently used to structure curriculum learning objectives and achieve the level of learning. The three domains and their respective levels are illustrated below:

Cognitive [C] (Knowledge-based)	**Affective [A]** (Emotion-based)	**Psychomotor [P]** (Action-based)
Remember	Receive	Imitate
Understand	Respond	Manipulate
Apply	Value	Precision
Analyze	Organize	Articulation

	Evaluate	Characterize	Naturalization
	Create	--- --- ---	--- --- ---

33 **Descriptions of Cognitive Domain (Anderson and Krathwohl'supdated Taxonomy in 2001):**

The **cognitive domain** involves the development of our mental skills and the acquisition of knowledge.

Category (Level)	Meaning	Keywords
Remember (C1)	Recognizing or recalling knowledge from memory. Remembering is when memory is used to produce or retrieve definitions, facts, or lists, or to recite previously learned information.	Define, describe, draw, find, identify, label, list, match, name, quote, recall, recite, tell, and write
Understand (C2)	Constructing meaning from different types of functions be they written or graphic messages or activities like interpreting, exemplifying, classifying, summarizing, inferring, comparing, or explaining.	Classify, compare, exemplify, conclude, demonstrate, discuss, explain, identify, illustrate, interpret, paraphrase, predict, and report
Apply (C3)	Carrying out or using a procedure through executing or implementing. Applying relates to or refers to situations where learned material is used through products like models, presentations, interviews, or simulations.	Apply, change, choose, compute, dramatize, implement, interview, prepare, produce, role play, select, show, transfer, and use
Analyze (C4)	Breaking materials or concepts into parts, determining how the parts relate to one another or how they interrelate, or how the parts relate to an overall structure or purpose. Mental actions included in this function are differentiating, organizing, and attributing, as well as being able to distinguish between the components or parts. When one is analyzing, he/she can illustrate this mental function by creating spreadsheets, surveys, charts, or diagrams, or graphic representations.	Analyze, characterize, classify, compare, contrast, debate, deconstruct, deduce, differentiate, discriminate, distinguish, examine, organize, outline, relate, research, separate, and structure
Evaluate (C5)	Making judgments based on criteria and standards through checking and critiquing. Critiques, recommendations, and reports are some of the products that can be created to demonstrate the processes of evaluation.	Appraise, argue, assess, choose, conclude, decide, evaluate, judge, justify, predict, prioritize, prove, rank, rate, select, Monitor.
Create (C6)	Putting elements together to formal coherent or functional whole reorganizing elements into a new pattern or structure through generating, planning, or producing. Creating requires users to put parts together in a new way, or synthesize parts into something new and different creating a new form or product. This process is the most difficult mental function in the new taxonomy.	Create, invent, compose, predict, plan, construct, design, propose, devise, and formulate

34	**Graduate Attributes (Program Learning Outcomes) of B.Sc. in CSE Program based on Washington Accord**

Program Learning Outcomes (PLOs) are brief statements that describe what students are expected to know and be able to do by the time of graduation. These relate to the knowledge, skills, and attitudes that students acquire throughout the entire course of a program. The students of the B.Sc. in CSE program are expected to achieve the following graduate attributes or program outcomes at the time of graduation:

PLO1. Apply knowledge of mathematics, natural science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PLO2. Identify, formulate, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics and natural and engineering sciences.

PLO3. Design solutions for complex engineering problems and design systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.

PLO4. Conduct investigations of complex problems using research-based knowledge and research methods that include the design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

PLO5. Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering problems, with an understanding of the limitations.

PLO6. Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems.

PLO7. Understand and evaluate the sustainability and impact of professional engineering work to solve complex engineering problems in societal and environmental contexts.

PLO8. Apply ethical principles and commit to professional ethics, responsibilities, and norms of engineering practice.

PLO9. Function effectively as individuals and members or leaders of diverse teams and in multidisciplinary settings.

PLO10. Communicate effectively on complex engineering activities with the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and convey and receive clear instructions.

PLO11. Demonstrate knowledge and understanding of engineering and management principles and apply them to their work as team members or leaders or entrepreneurs to manage projects in multidisciplinary environments.

PLO12. Recognize self-awareness to engage in a lifelong learning process to reflect in the broadest context of technological change.

35	**Knowledge Profile (K)**

K1: A systematic, theory-based understanding of the natural sciences applicable to the discipline.

K2: Conceptually-based mathematics, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modeling applicable to the discipline.

K3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

K4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline much is at the forefront of the discipline.

K5: Knowledge that supports engineering design in a practice area.

K6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

K7: Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the professional responsibility of an engineer to public safety; the impacts of engineering activity: economic, social, cultural, environmental and sustainability.

K8: Engagement with selected knowledge in the research literature of the discipline.

36	Range of Complex Engineering Problem Solving (P)
	Complex Engineering Problems have characteristic **P1** and several or all of **P2** to **P7**: **P1. Depth of knowledge required**: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8, which allows a fundamentals-based, first principles analytical approach **P2. Range of conflicting requirements**: Involve wide-ranging or conflicting technical, engineering and other issues **P3. Depth of analysis required**: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models **P4. Familiarity of issues**: Involve infrequently encountered issues **P5. Extent of applicable codes:** Are outside problems encompassed by standards and codes of practice for professional engineering **P6. Extent of stakeholder involvement and conflicting requirements**: Involve diverse groups of stakeholders with widely varying needs **P7. Interdependence**: Are high-level problems including many component parts or sub-problems

37	Range of Complex Engineering Activities (A)
	Attribute Complex activities means (engineering) activities or projects that have several or all of the following characteristics: **A1. Range of resources**: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies). **A2. Level of interaction**: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering, or other issues. **A3. Innovation**: Involve creative use of engineering principles and research-based knowledge in novel ways. **A4. Consequences for society and the environment**: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation. **A5. Familiarity**: Can extend beyond previous experiences by applying principles-based approaches.
38	**Code of Conduct**

	It is strongly suggested that students keep discipline in the classroom by attending class on time, listening to lectures attentively, and participating in discussions on the subject. To get class participation grades, students MUST attend the classes of the courses s/he registered for. Turn off his or her cell phone before entering a class or participating in class tests and exams. There are activities that are considered academic misconduct. One of them is plagiarism, which signifies the deliberate formal presentation or submission of works, phrases, texts, ideas, illustrations, or diagrams of others as one's own without proper citation. Another one is the use of unauthorized aids (including electronic devices), asking for assistance, or using illegal materials when preparing assignments or in examinations. In addition, copying from others' work, showing your work to others, and asking for answers are also considered academic misconduct. Penalties for involving academic misconduct include one or more of the following: a zero grade on the work produced, a failing grade in the course, suspension for one semester or more, and even expulsion from the university. On the university premises or at a university-sponsored program, students must abide by the Student Code of Conduct and other Rules and Regulations of BUBT, which are available on the BUBT website at https://www.bubt.edu.bd/Home/page_details/Office_of_the_Proctor.

39	**Social & Moral Values**		
	Our promises are based on the three cardinal principles: i. What we do believe. ii. What we do practice. iii. What we will promote. However, students are advised to undertake the following commitments for social and moral developments.		
	• To be punctual and attentive in classes; • To prioritize honesty & faith; • To ensure mutual respect; • To be always proactive; • To avoid conspiracy; • To be cooperative in learning; • To be sincere in class preparation;	• To avoid unfair means and plagiarism in exams, report writings and assignments; • To carry out assignments or keep other commitments timely; • To be motivated for asking question and encourage feedback; • Not to forget to switch-off the cellphone in a class;	• To follow the dress code and wearing ID card on campus; • To be decenton all aspects; • To be loyal and trust-worthy to the teachers and others; • Help keeping an eco-friendly environment in the campus.

Prepared by: **Checked by:** **Approved by:**

Nasrin Akter

Lecturer

Dept of CSE